

Single Variable Calculus

Personal Notes

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1 Derivatives, Slope, Velocity, and Rate of Change

Calculus links local change to global behavior. Its first local object is the derivative: geometrically, it is the slope of a tangent line; physically, it is an instantaneous rate such as velocity. Both interpretations come from the same limiting process.

1.1 Secant Slopes and the Derivative

Let $y = f(x)$ and fix a point $P = (x_0, f(x_0))$ on the graph. A nearby point $Q = (x_0 + \Delta x, f(x_0 + \Delta x))$ determines a *secant line*. Its slope is the *difference quotient*

$$\frac{\Delta f}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

As Q approaches P , the secant line may approach a limiting line: the tangent line at P .

Definition 1.1 [*Derivative*]. If the limit exists, the derivative of f at x_0 is

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

It is the slope of the tangent line to the graph of f at x_0 .

The tangent line should not be defined merely as a line that meets the graph at one point: a line can cross a curve once without reflecting its local direction. The limiting-secants definition captures the direction of the curve near the point.

Note. The limit concerns nonzero values of Δx approaching zero. Substituting $\Delta x = 0$ into the difference quotient gives the meaningless expression $0/0$; algebraic simplification must come before taking the limit.

1.2 Example: the Reciprocal Function

Take $f(x) = 1/x$ at a point $x_0 \neq 0$. For $\Delta x \neq 0$,

$$\begin{aligned} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} &= \frac{1}{\Delta x} \left(\frac{1}{x_0 + \Delta x} - \frac{1}{x_0} \right) \\ &= \frac{1}{\Delta x} \left(\frac{-\Delta x}{x_0(x_0 + \Delta x)} \right) \\ &= -\frac{1}{x_0(x_0 + \Delta x)}. \end{aligned}$$

The cancellation makes the limit visible:

$$f'(x_0) = -\frac{1}{x_0^2}.$$

The negative sign agrees with the graph: $1/x$ is decreasing on either side of the origin.

The tangent line through $(x_0, 1/x_0)$ is therefore

$$y - \frac{1}{x_0} = -\frac{1}{x_0^2}(x - x_0).$$

Setting $y = 0$ gives the x -intercept $2x_0$. By symmetry of $y = 1/x$ under exchanging x and y , the y -intercept is $2/x_0$. The triangle cut off by the coordinate axes therefore has the constant area $\frac{1}{2}|2x_0||2/x_0| = 2$.

1.3 Derivative Notation

Because several traditions developed alongside calculus, the derivative has several standard notations. Starting with

$$\Delta y = \Delta f = f(x_0 + \Delta x) - f(x_0),$$

the limiting relation can be written

$$\frac{\Delta y}{\Delta x} \longrightarrow \frac{dy}{dx} \quad \text{or} \quad \frac{\Delta f}{\Delta x} \longrightarrow f'(x_0).$$

Thus dy/dx , df/dx , f' , and Df all denote a derivative. The notation $f'(x_0)$ records the evaluation point explicitly; when using dy/dx , the point must be supplied by context.

1.4 The Power Rule from the Definition

For a positive integer n , let $f(x) = x^n$. The binomial expansion gives

$$(x + \Delta x)^n = x^n + nx^{n-1}\Delta x + (\Delta x)^2 R(x, \Delta x),$$

where R collects the remaining finitely many terms. Hence

$$\frac{(x + \Delta x)^n - x^n}{\Delta x} = nx^{n-1} + \Delta x R(x, \Delta x).$$

The second term vanishes as $\Delta x \rightarrow 0$, proving the *power rule* for monomials and, term by term, for polynomials:

$$\frac{d}{dx}x^n = nx^{n-1}.$$

1.5 Velocity as an Instantaneous Rate

The same difference quotient has a physical meaning. If $y(t)$ is the height of an object, then

$$\frac{\Delta y}{\Delta t}$$

is its average velocity over a time interval, while the limiting value $y'(t)$ is its *instantaneous velocity*.

For an object dropped from a height of 400 feet, the model

$$y(t) = 400 - 16t^2$$

places impact at $t = 5$ seconds because $y(5) = 0$. Its average downward speed over the full fall is $400/5 = 80$ feet per second. The instantaneous velocity is

$$y'(t) = -32t,$$

so at impact

$$y'(5) = -160 \text{ ft/s.}$$

The negative sign records direction: height is decreasing. Speed is the magnitude, 160 feet per second.

Note. The tangent-slope and velocity viewpoints are not separate definitions. Both are limits of average change over shorter and shorter intervals, which is why one derivative describes geometry and motion at once.

Source: [MIT 18.01 Single Variable Calculus](#), Fall 2006, Lecture 1, “Derivatives, Slope, Velocity, and Rate of Change,” taught by David Jerison.

2 Limits, Continuity, and Trigonometric Limits

The derivative turns an average rate of change into an instantaneous one by taking a limit. This lecture makes that limiting operation explicit, uses it to classify continuity and discontinuity, and establishes the two trigonometric limits needed to differentiate sine and cosine.

2.1 Rates of Change and Sensitivity

For related quantities x and $y = f(x)$, the quotient

$$\frac{\Delta y}{\Delta x}$$

measures average change over an interval. As $\Delta x \rightarrow 0$, it approaches the instantaneous rate dy/dx when the derivative exists. The same pattern appears in many settings:

$$\frac{dq}{dt} = \text{electric current}, \quad \frac{ds}{dt} = \text{velocity}, \quad \frac{dT}{dx} = \text{spatial temperature gradient}.$$

A derivative also measures *sensitivity*. If a measured quantity h determines an inferred quantity L , then $\Delta L/\Delta h$ describes how measurement error propagates over a finite change, while dL/dh is the local sensitivity. A large magnitude means that a small error in h can cause a much larger error in L .

2.2 Limits and Direct Substitution

Definition 2.1 [*Limit*]. We write

$$\lim_{x \rightarrow a} f(x) = L$$

when the values of $f(x)$ approach L as x approaches a through values $x \neq a$. The value $f(a)$ need not be used, or even defined.

Some limits are found by direct substitution. For example,

$$\lim_{x \rightarrow 3} \frac{x^2 + x}{x + 1} = \frac{3^2 + 3}{3 + 1} = 3.$$

This works because the expression is defined at $x = 3$ and its denominator stays nonzero nearby. In contrast, a derivative limit initially gives $0/0$ under direct substitution:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

The fraction is considered only for $x \neq a$; it must be simplified or estimated before the limit is taken.

2.3 Continuity and One-Sided Limits

Definition 2.2 [*Continuity*]. A function f is continuous at a when $f(a)$ is defined, the limit $\lim_{x \rightarrow a} f(x)$ exists, and

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Continuity says that the function's nearby values agree with its assigned value at the point. To distinguish the two directions of approach, the *right-hand limit*

$$\lim_{x \rightarrow a^+} f(x)$$

uses only $x > a$, while the *left-hand limit* $\lim_{x \rightarrow a^-} f(x)$ uses only $x < a$. A two-sided limit exists exactly when the two one-sided limits exist and agree.

These limits separate four common failures of continuity.

- A *removable discontinuity* occurs when the two one-sided limits agree in a finite value L , but $f(a)$ is missing or differs from L . Defining $f(a) = L$ repairs the discontinuity. The function $\sin x/x$ at $x = 0$ will be an example once its limit is known.
- A *jump discontinuity* occurs when both one-sided limits are finite but unequal. Changing only $f(a)$ cannot join the two sides.
- An *infinite discontinuity* occurs when values become unbounded on at least one side, as $1/x$ does near zero.
- An *oscillatory discontinuity* occurs when values keep fluctuating without approaching any finite or infinite limit, as $\sin(1/x)$ does near zero.

2.4 Reading a Derivative Graph

The graph of a function and the graph of its derivative generally look quite different. For $f(x) = 1/x$,

$$f'(x) = -\frac{1}{x^2}.$$

The original function changes sign across the origin, whereas its derivative is negative on both sides. The derivative graph records slope, not height.

Symmetry gives a useful structural relationship. An *odd function* satisfies $f(-x) = -f(x)$; differentiating shows that f' is even. An *even function* satisfies $f(-x) = f(x)$; differentiating shows that f' is odd, whenever the derivatives exist.

For another picture, consider

$$y(t) = 400 - 16t^2, \quad -5 \leq t \leq 5.$$

It models an object rising from ground level, reaching its maximum height at $t = 0$, and returning to the ground. Its velocity

$$y'(t) = -32t$$

is positive during ascent, zero at the top, and negative during descent. The sign of the derivative therefore identifies the direction of motion directly.

2.5 Two Trigonometric Limits

Angles must be measured in *radians*. For $0 < \theta < \pi/2$, comparison of a triangle and a unit-circle sector gives

$$\sin \theta < \theta < \tan \theta.$$

Dividing through and rearranging yields

$$\cos \theta < \frac{\sin \theta}{\theta} < 1.$$

Both outer quantities approach 1 as $\theta \rightarrow 0^+$. Since $\sin \theta / \theta$ is even, the same conclusion holds from the left:

$$\boxed{\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.}$$

The identity

$$1 - \cos \theta = \frac{(1 - \cos \theta)(1 + \cos \theta)}{1 + \cos \theta} = \frac{\sin^2 \theta}{1 + \cos \theta}$$

then gives

$$\frac{1 - \cos \theta}{\theta} = \frac{\sin \theta}{\theta} \frac{\sin \theta}{1 + \cos \theta} \rightarrow 1 \cdot 0,$$

so

$$\boxed{\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0.}$$

2.6 Differentiability Implies Continuity

Theorem 2.3 [*Differentiable functions are continuous*]. If f is differentiable at a , then f is continuous at a .

Proof. For $x \neq a$, factor the change in f as

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a).$$

As $x \rightarrow a$, differentiability makes the first factor approach $f'(a)$, while the second approaches 0. Their product therefore approaches 0, and hence $\lim_{x \rightarrow a} f(x) = f(a)$. $\square$

Note. The converse is false. The function $f(x) = |x|$ is continuous at zero but has different left- and right-hand slopes there, so it is not differentiable at zero.

Source: [MIT 18.01 Single Variable Calculus](#), Fall 2006, Lecture 2, “Limits, Continuity, and Trigonometric Limits,” taught by David Jerison.

3 Derivatives of Products, Quotients, Sine, and Cosine

The derivative definition can be evaluated separately for each new function, but a small collection of general formulas makes differentiation systematic. This lecture develops the linearity, trigonometric, product, and quotient rules directly from limits.

3.1 Linearity of the Derivative

Suppose u and v are differentiable functions of x , and let c be a constant. The *sum rule* and *constant-multiple rule* are

$$(u + v)' = u' + v', \quad (cu)' = cu'.$$

Both follow by splitting constants and sums inside the difference quotient. For the sum, for example,

$$\begin{aligned}(u + v)'(x) &= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x) + v(x + \Delta x) - u(x) - v(x)}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x) - u(x)}{\Delta x} + \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x) - v(x)}{\Delta x} \\&= u'(x) + v'(x).\end{aligned}$$

The calculation for cu is the same, with c factored outside the limit.

3.2 Derivatives of Sine and Cosine

Lecture 2 established the small-angle limits

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1, \quad \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0.$$

These formulas require angles in radians. Applying the sine addition identity to the derivative definition gives

$$\begin{aligned}\frac{d}{dx} \sin x &= \lim_{h \rightarrow 0} \frac{\sin(x + h) - \sin x}{h} \\&= \lim_{h \rightarrow 0} \left(\sin x \frac{\cos h - 1}{h} + \cos x \frac{\sin h}{h} \right) \\&= \cos x.\end{aligned}$$

A parallel calculation with $\cos(x + h) = \cos x \cos h - \sin x \sin h$ yields

$$\frac{d}{dx} \cos x = -\sin x.$$

The sign is geometrically consistent: near $x = 0$, cosine begins at its maximum and decreases to the right, while sine has positive unit slope.

3.3 The Product Rule

The derivative of a product is not generally the product of the derivatives. Instead, the rule is

Theorem 3.1 [*Product rule*]. If u and v are differentiable, then

$$(uv)' = u'v + uv'.$$

Proof. Insert and subtract $u(x+h)v(x)$ in the numerator of the difference quotient:

$$\begin{aligned}(uv)'(x) &= \lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x)v(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{u(x+h) - u(x)}{h} v(x) + u(x+h) \frac{v(x+h) - v(x)}{h} \right].\end{aligned}$$

Differentiability implies continuity, so $u(x+h) \rightarrow u(x)$. Taking the limits of the two terms gives $u'(x)v(x) + u(x)v'(x)$. $\square$

There is also an area interpretation. If a rectangle has side lengths u and v , changes Δu and Δv change its area by

$$(u + \Delta u)(v + \Delta v) - uv = v \Delta u + u \Delta v + \Delta u \Delta v.$$

The first two terms are the two strips added along the original sides; the last is the small corner rectangle. Dividing by Δx gives

$$\frac{\Delta(uv)}{\Delta x} = v \frac{\Delta u}{\Delta x} + u \frac{\Delta v}{\Delta x} + \frac{\Delta u}{\Delta x} \Delta v.$$

As $\Delta x \rightarrow 0$, continuity gives $\Delta v \rightarrow 0$, so the corner contribution vanishes and the two strip contributions become $vu' + uv'$.

3.4 The Quotient Rule

For $v(x) \neq 0$, the rule is

Theorem 3.2 [*Quotient rule*]. If u and v are differentiable and $v \neq 0$, then

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}.$$

To see the numerator's order and the squared denominator, write $\Delta u = u(x+h) - u(x)$ and $\Delta v = v(x+h) - v(x)$. Then

$$\frac{u + \Delta u}{v + \Delta v} - \frac{u}{v} = \frac{v \Delta u - u \Delta v}{(v + \Delta v)v}.$$

Divide by h :

$$\frac{1}{h} \left(\frac{u + \Delta u}{v + \Delta v} - \frac{u}{v} \right) = \frac{v(\Delta u/h) - u(\Delta v/h)}{(v + \Delta v)v}.$$

As $h \rightarrow 0$, the two difference quotients approach u' and v' , while $\Delta v \rightarrow 0$. Therefore the right-hand side approaches $(u'v - uv')/v^2$.

Note. The product and quotient rules each contain two contributions because both inputs vary. A useful memory check is that the product rule is symmetric in u and v , whereas the quotient rule is not: its numerator is “derivative of the top times the bottom, minus the top times derivative of the bottom.”

Source: [MIT 18.01 Single Variable Calculus](#), Fall 2006, Lecture 3, “Derivatives of Products, Quotients, Sine, and Cosine,” presented by Kobi Kremnizer.

4 Chain Rule, and Higher Derivatives

The rules from Lecture 3 differentiate sums, products, and quotients. The next operation is *composition*: if $x = g(t)$ and $y = f(x)$, then

$$y = f(g(t)) = (f \circ g)(t).$$

The chain rule explains how the rate of the outer function combines with the rate of the inner function.

4.1 The Chain Rule

Write the changes in the three variables as Δt , Δx , and Δy . When the relevant quotients are defined,

$$\frac{\Delta y}{\Delta t} = \frac{\Delta y}{\Delta x} \frac{\Delta x}{\Delta t}.$$

As $\Delta t \rightarrow 0$, continuity of g gives $\Delta x \rightarrow 0$. This suggests the derivative formula below.

Theorem 4.1 [Chain rule]. *If g is differentiable at t and f is differentiable at $g(t)$, then*

$$\frac{d}{dt} f(g(t)) = f'(g(t))g'(t).$$

Equivalently, if $y = f(x)$ and $x = g(t)$, then

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}.$$

Proof. Set $x_0 = g(t)$ and let $\Delta x = g(t + h) - g(t)$. Differentiability of f at x_0 means that

$$f(x_0 + \Delta x) - f(x_0) = f'(x_0)\Delta x + r(\Delta x)\Delta x,$$

where $r(\Delta x) \rightarrow 0$ as $\Delta x \rightarrow 0$. Dividing by h gives

$$\frac{f(g(t + h)) - f(g(t))}{h} = (f'(x_0) + r(\Delta x)) \frac{g(t + h) - g(t)}{h}.$$

Letting $h \rightarrow 0$ yields $f'(g(t))g'(t)$. □

Example: Let $f(x) = \sin x$ and $g(t) = t^2$. Then

$$\frac{d}{dt} \sin(t^2) = \cos(t^2) \cdot 2t = 2t \cos(t^2).$$

The outer sine contributes the cosine factor, evaluated at t^2 , and the inner square contributes $2t$.

Order matters in composition. With $f(x) = \sin x$ and $g(x) = x^2$,

$$(f \circ g)(x) = \sin(x^2), \quad (g \circ f)(x) = \sin^2 x.$$

Thus $f \circ g$ and $g \circ f$ need not be equal.

4.2 Two More Chain-Rule Examples

For $y = \cos(1/x)$ with $x \neq 0$, set $u = 1/x$. Since

$$\frac{dy}{du} = -\sin u, \quad \frac{du}{dx} = -\frac{1}{x^2},$$

the two negative signs cancel:

$$\frac{d}{dx} \cos\left(\frac{1}{x}\right) = \frac{\sin(1/x)}{x^2}.$$

The chain rule also extends the power rule to negative integer exponents. For a positive integer n ,

$$x^{-n} = \left(\frac{1}{x}\right)^n.$$

Therefore

$$\begin{aligned} \frac{d}{dx} x^{-n} &= n \left(\frac{1}{x}\right)^{n-1} \left(-\frac{1}{x^2}\right) \\ &= -n x^{-n-1}. \end{aligned}$$

Writing the same function as $1/x^n$ and applying the reciprocal form of the quotient rule produces the same result.

4.3 Higher Derivatives

A *higher derivative* is obtained by differentiating a derivative again. The second derivative of f is the derivative of f' , and the process can be repeated:

$$f'' = (f')', \quad f''' = (f'')', \quad f^{(n)} = (f^{(n-1)})'.$$

Common notations are

$$f'(x) = Df = \frac{df}{dx}, \quad f''(x) = D^2f = \frac{d^2f}{dx^2}, \quad f^{(n)}(x) = D^n f = \frac{d^n f}{dx^n}.$$

The exponent on D counts repeated differentiation; it is not an ordinary power of the value Df .

Repeatedly differentiating x^n reveals a product of descending integers:

$$\begin{aligned} Dx &= 1, \\ D^2x^2 &= D(2x) = 2, \\ D^3x^3 &= D^2(3x^2) = D(6x) = 6, \\ D^4x^4 &= D^3(4x^3) = D^2(12x^2) = D(24x) = 24. \end{aligned}$$

The *factorial* of a positive integer n is

$$n! = n(n-1) \cdots 2 \cdot 1.$$

The observed pattern is therefore $D^n x^n = n!$.

Proposition 4.2. *For every positive integer n ,*

$$D^n x^n = n!.$$

Proof. For $n = 1$, $Dx = 1 = 1!$. Suppose $D^n x^n = n!$. Then the power rule and linearity give

$$\begin{aligned} D^{n+1}x^{n+1} &= D^n(Dx^{n+1}) \\ &= D^n((n+1)x^n) \\ &= (n+1)D^n x^n \\ &= (n+1)n! = (n+1)!. \end{aligned}$$

Thus the result follows by induction. □

Note. The chain rule and higher derivatives answer different questions. The chain rule differentiates one composition; higher-derivative notation records how many times differentiation has been repeated.

Source: [MIT 18.01 Single Variable Calculus](#), Fall 2006, Lecture 4, “Chain Rule, and Higher Derivatives.”